

Bosch, Infineon, Nordic Semiconductor, NXP Semiconductors, and Qualcomm, has been established in Munich, Germany, to promote the adoption of RISC-V. The company, formed on December 22, 2023, aims to provide reference architectures and develop solutions for industries such as automotive, mobile, and IoT. Led by CEO Alexander Kocher, Quintauris plans to work closely with the RISC-V community to accelerate development and enhance the semiconductor ecosystem's resilience.

RISC-V SWOT analysis

Strengths.

- RISC-V is the first new CPU IP architecture in 15 years to gain significant market presence. The ecosystem is growing, with multiple companies developing products for all SoC design requirements, including major players like Synopsys.
- The core ISA is fixed, but customisation through ISA extensions is possible and continually evolving. Low power consumption and increasing performance metrics are attractive to SoC designers for crafting powerful silicon solutions. Performance advances are closing the gap between RISC-V and the highest-performing CPU cores from competitors.
- The RISC-V community is diverse, leading to the rapid emergence of new ideas and concepts. Competition fostered by innovation at this fast pace is propelling market growth. Companies are developing CPU IP cores in parallel and focusing on a full range of performance metrics. Support from software

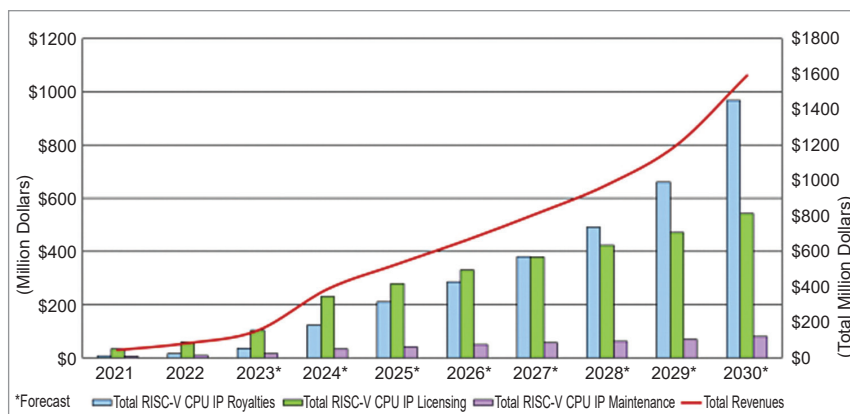


Fig. 5: Total RISC-V CPU IP market revenues, 2021-2030 (Source: The SHD Group, January 2024)

companies and EDA tools is increasing as the market grows.

Weaknesses.

- As a developing ecosystem, RISC-V has its weaknesses. Many small companies developing products have little to no track record in the market. Many of these companies are start-ups that must continually raise money for funding before revenue streams start, creating the possibility of high turnover and turmoil.
- There is no overall cohesion in product development between companies, leading to some duplication of efforts. The ability to customise ISA extensions could lead to ISA fractionalisation if not managed by an entity like RISC-V International.
- It will take time for the ecosystem to flesh out completely, while competitive ecosystems are already in place. Attracting software developer support is critical for success.

Opportunities.

- New, larger systems manufacturers are looking at RISC-V, with several already having plans for adoption. The heterogeneous

nature of the SoC design landscape today presents a great opportunity for RISC-V to be adopted alongside other competing CPU cores.

- RISC-V's ability to customise ISA extensions provides great flexibility for designers, especially for domain-specific solutions. Its open nature allows for constant innovation, with new ideas surfacing all the time.

- Architectural exploration and development of solutions for AI silicon favour the adoption of RISC-V CPU cores due to their low power envelope and rising performance metrics.

- The relatively small size of RISC-V CPU IP cores allows for efficient die area usage, making them attractive to SoC designers developing SoCs with large core counts.

Threats.

- Concern over the open standard nature of RISC-V and confusion between specifications and implementations may lead to increased government oversight and overregulation. Overregulation may curtail collaboration and hamper innovation.

- Market enthusiasm and momentum may dwindle over time before the ecosystem can be completely fleshed out. EDA vendors may shift focus away from RISC-V to favour other emerging technologies over time.

Major wins for RISC-V

- Launch of RISC-V Software Ecosystem (RISE) and adoption by prominent CPU IP vendors
- Intel joining RISC-V International and collaboration between Qualcomm and Google
- Increased availability of EDA tools for RISC-V and enhancements in CPU cores
- Introduction of RISC-V based MCUs, MPUs, and SoCs by Renesas, Meta, and Alibaba
- Synopsys transitioning ARC CPU IP cores to RISC-V ISA